Vincent Lin

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Education

Rochester Institute of Technology, Rochester, NY Aug 2022 – May 2027
GPA: 3.41 | **Awards**: Uncommon Hacks 2026 Social Track 1st Place, Dean's List, AP Scholar
B.S. in Mechatronics Engineering Technology May 2026
B.S. in Electrical Engineering Technology Expected May 2027

Experience

Hardware Design Engineer Jun 2026 – Aug 2026
LoomLogic Inc *New York, NY*
- Led the end-to-end redesign of edge-AI textiles classifier, advancing from R&D to production-ready mechanical design in 6 weeks
- Modeled the hardware workstation assembly in SolidWorks by CADing and integrating 4 MVP electromechanical subsystems
- Validated system design against VOC and functional requirements by conducting beam deflection and lighting/FOV analysis
- Developed complete Excel BOM in 3 weeks under $3000 by applying DFMA to standardize hardware and simplify assembly
- Finalized production hand-off by commissioning the workstation and documenting assembly in an installation manual

Robotics Automation Field Engineer Sep 2025 – Dec 2025
Retiina *USPS Atlanta RPDC*
- Operated and optimized USPS automated parcel sortation system, supporting 800,000+ daily package throughput
- Led the Operational Excellence team in a DOE-style motor analysis project, reducing system downtime to 8%
- Designed IoT data collection system to gauge motor noise, temperature, and vibration for predictive maintenance

Design and Manufacturing Intern May 2025 – Aug 2025
Electronic Hardware Corporation *Ronkonkoma, NY*
- Modeled and prototyped injection-moldable control knobs in Autodesk Inventor for Defense and Aerospace OEMs
- Developed 100+ new GD&T-compliant drawings in AutoCAD and managed entire drawing database in Macola
- Created official manufacturing work orders for custom and catalog parts based on in-house manufacturing processes

Makerspace Lab Assistant (Part-Time) Jan 2024 – Present
RIT SHED *Rochester, NY*
- Trained over 2000 students in learning to use FDM/SLA Printers, Laser Cutters, Soldering and Machine Shop Tools

Development Engineer Intern Jun 2024 – Aug 2024
OPmobility, Plastic Omnium KK *Tokyo, Japan*
- Assisted validation of SCR emission control system using Vector CANape and Python, completing unit testing in 8 weeks

Projects

LifeStory – Memory Record Player | *UncommonHacks '26 1st Place* devpost.com/software/lifestory
- Developed a photo-based record player that turns physical photos into memory playback for Alzheimer's patients
- Built computer vision pipeline using ESP32-S3 CAM, DINOv2 embeddings, ORB matching, and RANSAC homography
- Tuned hybrid DINOv2 + ORB matching algorithm through multiple edge cases, significantly reducing false positives

Automated SVG Sisyphus Table Mar 2026 – Present
- Built SVG input to Polar G-code converter using NumPy and coordinate geometry to drive a Polar CNC sand table
- Developed C++ firmware on Nano to drive NEMA17's and used PySerial as data bridge from Python terminal
- Designed and verified electrical schematic for complete motor control system and simulation in CirkitDesigner IDE

PACK-E – Autonomous Object Detection & Retrieval Robot Mar 2026 – Present
- Built real-time object detection system using fine-tuned FOMO model running TensorFlow Lite Micro on ESP32-S3
- Developed bare-metal C firmware for MSP432 motor control, ultrasonic sensing, and inference feedback via UART
- Created navigation logic by computing error between centroid location and frame center using local FOMO output

Automated Blast Gates | *Electrical/Embedded Systems Lead* Jan 2026 – May 2026
- Optimized vacuum flow and shop safety by designing, manufacturing, and integrating "smart" blast gates with the Wood Shop
- Developed autonomous closed-loop positioning circuit using L298N integrated with Nano logic and linear actuator feedback
- Established ESP32-Nano broadcast architecture via 8N1 UART and devised flow balancing equation for local computation
- Validated reliability over 5,000 actuation cycles against functional spec by using DFMEA results to conduct endurance testing

Skills

Software: C, C++, Python, Git, OpenCV, PyTorch, Linux, MATLAB, ABB RAPID Programming, Minitab, Microsoft Office Suite
Robotics/Embedded: ROS2/micro-ROS, ARM-Cortex M4, ESP32-S3, NI Multisim, Arduino, PlatformIO, VHDL, ABB RobotStudio, Allen-Bradley PLC, Studio 5000, TinyML, Edge Impulse, UART/I2C/SPI, DE0-CV FPGA, Intel Quartus Prime
Mechanical: Autodesk Inventor, SolidWorks, Fusion 360, AutoCAD, Laser Cutting, 3D Printing, GD&T, Soldering, Machine Shop Tools, CNC Mill, Lean Six Sigma, Statistical Design of Experiments, DFMEA, MS Project

Certifications

Lean Six Sigma – Yellow Belt | Certified SolidWorks Associate – Mechanical Design | Autodesk Inventor Certified User